

3-Mark Questions & Answers — Data Structures Module 6: Sorting & Searching Techniques

Q1. Explain the four common Hash Functions.

- 1. **Division Method:** $h(k) = k \bmod m$ (where m is prime).
- 2. **Mid-Square Method:** Square k^2 and extract middle r digits.
- 3. **Folding Method:** Divide key into parts and sum them.
- 4. **Multiplication Method:** $h(k) = \lfloor m(kA \bmod 1) \rfloor$.

Q2. Compare Open Addressing and Separate Chaining collision resolution.

Feature	Open Addressing	Separate Chaining
Storage	Inside table array slots	Array of linked lists
Load Factor	$\alpha \leq 1$	$\alpha > 1$ possible
Deletion	Requires DELETED markers	Simple linked list node deletion
Cache Locality	Excellent	Poor (pointer chasing)